

MTE 204 – Numerical Methods

Tutorial #1 Questions

Question 1:

The volume $\bar{V}$ of liquid in a spherical tank of radius R is related to the depth of the liquid h by the following equation.

$$\bar{V} = \frac{\pi h^2(3R - h)}{3}$$

For a spherical tank with a radius of 1 m and containing 0.5 m^3 of liquid, find the depth of the liquid in the tank. Use fixed point iteration for this problem.

Question 2:

Determine the largest real root of $f(x) = x^3 - 6x^2 + 11x - 6.1$ using the following two methods

- The graphical method
- The Newton-Raphson method. Perform 3 iterations using $x_0 = 3.5$

Question 3:

The velocity of a falling parachutist is given by $v = \frac{gm}{c} \left(1 - e^{-\left(\frac{c}{m}\right)t} \right)$. Using the bisection method, determine the drag coefficient needed such that an 80 kg parachutist has a velocity of 35 m/s after 3.9 s of freefall. Assume gravity is 9.81 m/s^2 . Use an initial bracket of [3,5] and iterate until approximate relative error falls below 5%. Perform an error check by substituting your final answer into the initial equation.